



ELSEVIER

8 July 1996

PHYSICS LETTERS A

Physics Letters A 217 (1996) 73–77

Coherent states for isospectral Hamiltonians

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Received 28 March 1996; accepted for publication 18 April 1996

Communicated by P.R. Holland

Abstract

We give a construction of coherent states for strictly isospectral Hamiltonians by exploiting the fact that these are related by a unitary transformation, and hence the corresponding coherent states must be related by the same unitary transformation. We give the explicit structure of such a unitary transformation in terms of the eigenvalues and eigenfunctions of a given factorizable Hamiltonian. As an illustration, we discuss the example of strictly isospectral one-dimensional harmonic oscillator Hamiltonians and the associated coherent states, and point out inadequacies in recent constructions of such coherent states in the literature.

PACS: 03.65.Fd; 02.30.+b

In recent years extensive work has been done on various aspects of coherent states [1,2]. The interest in coherent states is largely due to the fact that they provide an alternative set of basis vectors (non-orthogonal, overcomplete), they label the phase space of the system, and in some cases (the harmonic oscillator being the best known example) they have minimal fluctuations (allowed by the Heisenberg uncertainty principle) in the canonically conjugate variables, and hence are closest to the classical states. In recent years, there has been a lot of interest in constructing coherent states for potentials other than the harmonic oscillator, for example the Morse potential [3], the Coulomb potential [4], etc.

Very recently, coherent states have also been constructed [5,6] for the family of Hamiltonians which are strictly isospectral to the harmonic oscillator

Hamiltonian [7,8]. Various approaches exist in the literature for the construction of such isospectral families, for example the factorization method [9], the Gelfand–Levitan method [10] and the approach of supersymmetry (SUSY) quantum mechanics (for a recent review see Ref. [11]), and all of them are essentially equivalent [12–14].

In this letter we address the following question: Given a Hamiltonian whose coherent states are known, how does one construct coherent states for the corresponding strictly isospectral Hamiltonians? We make the simple, but crucial, observation that any two strictly isospectral Hamiltonians are related by a unitary transformation and as a consequence the corresponding coherent states must be related by the same unitary transformation. We give an explicit construction of the unitary operators. We would like to remark that although the statement that isospectral Hamiltonians are unitarily related need hardly be emphasized, it is however important to note two points: (i) that

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we are dealing with isospectral families of Hamiltonians which are all Schrödinger operators (i.e., they describe a particle moving in a potential), and (ii) that only for *factorizable* Hamiltonians (Schrödinger operators) can one explicitly construct such a unitary transformation. As an illustration of our general result, we discuss in detail the specific example of the strictly isospectral oscillator family. We also show that recent constructions [5,6] of the coherent states for this family of potentials are not satisfactory. This is because these coherent states do not go over to the harmonic oscillator coherent states in the appropriate limit. In contrast, we show that our coherent states do.

Consider the operators ($\hbar = m = 1$)

$$\begin{aligned} a &= \frac{1}{\sqrt{2}} \left(\frac{d}{dx} + W(x) \right), \\ a^\dagger &= \frac{1}{\sqrt{2}} \left(-\frac{d}{dx} + W(x) \right), \end{aligned} \quad (1)$$

where $W(x)$ is an arbitrary function of x . It is well known that the Hamiltonians $a^\dagger a$ and $aa^\dagger$ are SUSY partners. As a consequence, eigenvalues, eigenfunctions and S -matrices of the two Hamiltonians are related, and if the eigenfunctions of $a^\dagger a$ can be solved for, then the eigenfunctions of $aa^\dagger$ can be obtained in terms of those of $a^\dagger a$.

Further, it is also well known that one can construct a family of strictly isospectral Hamiltonians $b^\dagger b$. Consider the operators

$$\begin{aligned} b &= \frac{1}{\sqrt{2}} \left(\frac{d}{dx} + \hat{W}(x) \right), \\ b^\dagger &= \frac{1}{\sqrt{2}} \left(-\frac{d}{dx} + \hat{W}(x) \right), \end{aligned} \quad (2)$$

such that

$$bb^\dagger = aa^\dagger. \quad (3)$$

The condition (3) leads to a Riccati equation which can be solved to give

$$\hat{W}(x) = W(x) + \phi_\lambda(x), \quad (4)$$

where

$$\phi_\lambda(x) = \psi_0^2(x) \left(\lambda + \int_{-\infty}^x dy \psi_0^2(y) \right)^{-1}, \quad (5)$$

where λ is a real number not lying in the closed interval $[-1, 0]$ and $\psi_0(x)$ is the normalized ground state eigenfunction of the Hamiltonian $a^\dagger a$.

The eigenstates $|\theta_n\rangle$ of the strictly isospectral family of Hamiltonians $H_\lambda = b^\dagger b + E_0$, (E_0 being the lowest energy eigenvalue) can be obtained in terms of the eigenstates $|\psi_n\rangle$ of $H = a^\dagger a + E_0$, by noting the fact that

$$(b^\dagger b)b^\dagger a = b^\dagger a(a^\dagger a). \quad (6)$$

Hence the normalized eigenstates of H_λ are

$$|\theta_n\rangle = \frac{1}{E_n - E_0} b^\dagger a |\psi_n\rangle, \quad n = 1, 2, \dots \quad (7)$$

The ground state of H_λ is determined from the condition $b|\theta_0\rangle = 0$. Hence the normalized eigenfunctions of H_λ in the coordinate representation are given by

$$\begin{aligned} \theta_0(x) &= \sqrt{\lambda(\lambda+1)} \left(\lambda + \int_{-\infty}^x dy \psi_0^2(y) \right)^{-1} \psi_0(x), \\ \theta_n(x) &= \psi_n(x) \\ &\quad + \frac{1}{2(E_n - E_0)} \phi_\lambda(x) \left(\frac{d}{dx} + W(x) \right) \psi_n(x), \\ n &= 1, 2, \dots \end{aligned} \quad (8)$$

Let us now show that the Hamiltonians $H = a^\dagger a + E_0$ and $H_\lambda = b^\dagger b + E_0$ are related by a unitary transformation, a fact which is not immediately obvious from the structure of the operators b and $b^\dagger$. The condition (3) implies that if one writes

$$b = aU^\dagger, \quad b^\dagger = Ua^\dagger, \quad (9)$$

then

$$U^\dagger U = 1. \quad (10)$$

We now define the operators

$$A = UaU^\dagger, \quad A^\dagger = Ua^\dagger U^\dagger, \quad (11)$$

so that $b^\dagger b = A^\dagger A = Ua^\dagger a U^\dagger$, and hence we have the relation

$$H_\lambda = UH U^\dagger. \quad (12)$$

Further, the fact that the Hamiltonians H and H_λ are isospectral and diagonal in the orthonormal bases $|\psi_n\rangle$ and $|\theta_n\rangle$ respectively, implies that

$$E_n = \langle \theta_n | H_\lambda | \theta_n \rangle = \langle \psi_n | H | \psi_n \rangle, \quad n = 0, 1, \dots \quad (13)$$

Hence, on using the relation (12) one has the equality (upto a phase factor that can be taken to be unity), i.e.

$$U^\dagger |\theta_n\rangle = |\psi_n\rangle. \quad (14)$$

The fact that the sets $|\psi_n\rangle$ and $|\theta_n\rangle$ are orthonormal implies that

$$\langle \psi_n | \psi_m \rangle = \langle \theta_n | U U^\dagger | \theta_m \rangle = \delta_{nm}, \quad (15)$$

and hence

$$U U^\dagger = 1. \quad (16)$$

Thus we have shown that the strictly isospectral Hamiltonians H and H_λ and also their respective eigenfunctions are related by the unitary transformation U defined by Eq. (9).

One way of defining coherent states for an arbitrary Hamiltonian is based on the dynamical symmetry group of the Hamiltonian [2]. Let G be the symmetry group of the Hamiltonian H . Hence one can express H as a linear combination of the generators J_i of the Lie algebra of G . Thus

$$H = \sum_i d_i J_i, \quad (17)$$

where d_i are complex numbers and J_i are closed under the commutation relations

$$[J_i, J_j] = c_{ij}^k J_k. \quad (18)$$

Now, since H_λ is unitarily related to H , one has

$$H_\lambda = \sum_i d_i \tilde{J}_i, \quad \tilde{J}_i = U J_i U^\dagger, \quad (19)$$

where $\tilde{J}_i$ obey the same Lie algebra (18) as do J_i , since the structure constants c_{ij}^k do not change under a unitary transformation. Thus H_λ has the same symmetry group as that of H . Let $D(z)$ be the element belonging to the coset space of G with respect to its maximal stability subgroup. Note that $D(z)$ is an operator function of the generators J_i . If the Perelomov coherent states associated with H and H_λ are defined respectively as

$$|z\rangle = D(z) |\psi_0\rangle, \quad |z; \lambda\rangle = D_\lambda(z) |\theta_0\rangle, \quad (20)$$

then, as a consequence of Eq. (19), one has

$$D_\lambda(z) = U D(z) U^\dagger, \quad (21)$$

and hence it follows from Eq. (14) that

$$|z; \lambda\rangle = U |z\rangle. \quad (22)$$

Thus we have shown that the Perelomov coherent states associated with the isospectral Hamiltonians H and H_λ are unitarily related. We would like to remark that this result is true no matter what definition one uses for the construction of the coherent states.

The explicit structure of the unitary operator U can easily be obtained. Expanding U in the complete set of eigenstates of H and using Eq. (14), we have

$$U = \sum_{n,m=0}^{\infty} U_{nm} |\psi_n\rangle \langle \psi_m|, \quad U_{nm} = \langle \psi_n | \theta_m \rangle. \quad (23)$$

Using Eq. (8) one can write down explicit expressions for the matrix elements U_{nm} . Thus

$$\begin{aligned} U_{n0} &= \int dx \sqrt{\lambda(\lambda+1)} \left(\lambda + \int_{-\infty}^x dy \psi_0^2(y) \right)^{-1} \\ &\quad \times \psi_n^*(x) \psi_0(x), \\ U_{n,m+1} &= \delta_{n,m+1} + \frac{1}{2(E_{m+1} - E_0)} \\ &\quad \times \int dx \psi_n^*(x) \phi_\lambda(x) \left(\frac{d}{dx} + W(x) \right) \psi_{m+1}(x), \\ n, m &= 0, 1, \dots \end{aligned} \quad (24)$$

It is simple to see from Eqs. (5) and (24) that in the limit $|\lambda| \rightarrow \infty$ one has $\phi_\lambda(x) \rightarrow 0$ and hence $U^\dagger, U \rightarrow 1$. As a result $H_\lambda \rightarrow H$, and hence the eigenstates as well as the coherent states associated with H_λ reduce to those of H in this limit.

We next consider in detail the specific example of the strictly isospectral oscillator Hamiltonians. Our motivation is two-fold: (i) to illustrate the general arguments made in the foregoing, and (ii) to show that constructions in the literature [5–7] of annihilation and creation operators and of coherent states associated with this family are unsatisfactory.

The harmonic oscillator is described by the Hamiltonian $H = a^\dagger a + \frac{1}{2}$ ($\hbar = m = \omega = 1$), where

$$\begin{aligned} a &= \frac{1}{\sqrt{2}} \left(\frac{d}{dx} + x \right), \\ a^\dagger &= \frac{1}{\sqrt{2}} \left(-\frac{d}{dx} + x \right), \end{aligned} \quad (25)$$

where the annihilation and creation operators a and $a^\dagger$ obey the usual commutation relations $[a, a^\dagger] = 1$. The isospectral oscillator family is then described by the Hamiltonian H_λ (see Eq. (12)),

$$H_\lambda = A^\dagger A + \frac{1}{2}, \quad A = UaU^\dagger, \quad A^\dagger = Ua^\dagger U^\dagger. \quad (26)$$

The structure of the unitary operator U (see Eq. (24)) in this case will depend on the eigenvalues and eigenfunctions of the harmonic oscillator H . Note that A and $A^\dagger$ obey the same commutation relations as do a and $a^\dagger$, i.e., $[A, A^\dagger] = 1$. Thus A and $A^\dagger$ are indeed the annihilation and creation operators associated with the isospectral Hamiltonian H_λ . Let us define new canonically conjugate operators, namely

$$\hat{X} = \frac{1}{\sqrt{2}}(A^\dagger + A), \quad \hat{P} = \frac{i}{\sqrt{2}}(A^\dagger - A),$$

$$[\hat{X}, \hat{P}] = i, \quad (27)$$

so that the Hamiltonian H_λ can be written in the form

$$H_\lambda = \frac{1}{2}(\hat{P}^2 + \hat{X}^2). \quad (28)$$

Note that $\hat{X}$ and $\hat{P}$ are related to the position and momentum operators $\hat{x}$ and $\hat{p}$ of the harmonic oscillator by the unitary transformation as in Eq. (26). Thus the family of isospectral oscillators can be viewed as harmonic oscillators but expressed in terms of appropriately transformed position and momentum operators. As a consequence, the coherent states associated with the isospectral oscillator family $|z; \lambda\rangle$ may be defined as eigenstates of the annihilation operator A , namely

$$A|z; \lambda\rangle = z|z; \lambda\rangle, \quad (29)$$

where the eigenvalue z is λ -independent, or equivalently, in the Perelomov sense as the displaced ground state, namely

$$|z; \lambda\rangle = D_\lambda(z) |\theta_0\rangle,$$

$$D_\lambda(z) = \exp(zA^\dagger - z^*A), \quad (30)$$

or, equivalently, as the state which has the minimum uncertainty product

$$\Delta X \Delta P = \frac{1}{2}, \quad (31)$$

(note that we have chosen $\hbar = m = \omega = 1$) with the uncertainties in X and P being equal. It must be noted

that these coherent states are not minimum uncertainty states with respect to the position and momentum of the particle, viz., x and p . As is well known, only the Gaussian states minimize the product $\Delta x \Delta p$. As argued more generally in the foregoing, the coherent states of the isospectral oscillator $|z; \lambda\rangle$ are related to the harmonic oscillator coherent states by a unitary transformation as in Eq. (26).

One may also define a more general state which minimizes the uncertainty product $\Delta X \Delta P$, in analogy with the squeezed coherent state [15] of the usual harmonic oscillator, as follows,

$$|\xi, z; \lambda\rangle = S_\lambda(\xi) D_\lambda(z) |\theta_0\rangle,$$

$$S_\lambda(\xi) = \exp\left[\frac{1}{2}\xi(A^\dagger)^2 - \frac{1}{2}\xi^* A^2\right], \quad (32)$$

where the displacement operator $D_\lambda(z)$ is as defined in Eq. (30). Note that in the state $|\xi, z; \lambda\rangle$ the uncertainties ΔX and ΔP are unequal while the product is one-half. As a consequence of Eq. (26) it follows that the state $|\xi, z; \lambda\rangle$ is related to the squeezed coherent state of the harmonic oscillator by a unitary transformation as in Eq. (26).

There have been some attempts in the literature [5–7] at the construction of a set of annihilation and creation operators, and of coherent states, associated with the isospectral oscillator family. However, these constructions have certain unsatisfactory features as we will show in the following: For example in Ref. [7], annihilation and creation operators ($A_M, A_M^\dagger$) are constructed. However they do not connect the ground state $|\theta_0\rangle$ to the excited states $|\theta_n\rangle$ ($n \geq 1$), they do not commute to an identity, and they do not reduce to the oscillator operators ($a, a^\dagger$) in the limit $|\lambda| \rightarrow \infty$ but instead reduce to $(a^\dagger a^2, (a^\dagger)^2 a)$. In Ref. [5], coherent states are constructed as the eigenstates of the annihilation operator A_M of Ref. [7] and these consequently do not reduce to the harmonic oscillator coherent states in the limit $|\lambda| \rightarrow \infty$. More recently, in Ref. [6], a family of pairs of annihilation and creation operators are constructed which obey a “distorted” Heisenberg algebra. Neither do these operators reduce to the oscillator operators ($a, a^\dagger$) in the limit $|\lambda| \rightarrow \infty$, but instead they reduce to some very complicated operators. Consequently, the coherent states that are constructed, either as eigenstates of these annihilation operators, or as states generated by the action of displacement operators (constructed out of the family of

annihilation–creation operators) on an extremal state (with respect to these annihilation operators) of the isospectral Hamiltonian, do not reduce to the harmonic oscillator coherent states in the limit $|\lambda| \rightarrow \infty$.

We would like to emphasize that unlike Ref. [7], our $(A, A^\dagger)$ defined by Eq. (27) are the correct set of annihilation and creation operators for the isospectral oscillator family, they act on *all* the eigenstates of the Hamiltonian H_λ , they commute to an identity, and they reduce to $(a, a^\dagger)$ in the limit $|\lambda| \rightarrow \infty$. Further, the coherent states associated with H_λ that we have constructed, also reduce, unlike in Refs. [5,6], to the harmonic oscillator coherent states in this limit. Moreover, unlike in Refs. [5,6], we get the same coherent states for the isospectral oscillator family, no matter what definition of the coherent states is used.

It may be noted that the coherent states $|z; \lambda\rangle$ associated with the isospectral family of Hamiltonians H_λ possess all the properties of the usual coherent states $|z\rangle$ associated with the Hamiltonian H , such as non-orthogonality, overcompleteness, etc., as these properties are invariant under a unitary transformation.

In conclusion, we have given a construction of the coherent states associated with the strictly isospectral family of Hamiltonians by exploiting the fact that these are related by a unitary transformation, and hence the corresponding coherent states are related by the same unitary transformation. We have given an explicit construction of the unitary transformation. We have discussed in detail the example of the isospectral oscillator family and shown that constructions of annihilation and creation operators and of coherent states for this family in the literature are unsatisfactory. We would like to remark that the conclusions of this letter are valid even in the case of n -parameter isospectral families of Hamiltonians [16].

MSK gratefully acknowledges discussions with Professor V. Srinivasan.

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